

# Kevin Wei

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## TUTORIAL / EDUCATION

**University of Pennsylvania, School of Engineering and Applied Science**, Philadelphia, PA

May 2028

*Bachelor of Science in Engineering (BSE) in Digital Media Design (GPA: 3.81/4.00)*

*Master of Science in Engineering (MSE) in Computer Graphics & Game Technology (GPA: 3.85/4.00)*

- *Relevant Coursework:* Data Structures & Algorithms; Computer Systems & Architecture; Discrete Math; Algorithmic Game Theory; Partial Differential Equations; Linear Algebra; Computer Animation; Interactive Computer Graphics; Procedural Computer Graphics; Advanced Rendering; 3D Game Development in Unreal
- *Activities:* UPGRADE; Penn SIGGRAPH; Penn Spark; Penn Lion Dancing; Content Manager @ Penn Review

## EXPERIENCE POINTS

**Lunar Simulation Internship** | *National Aeronautics and Space Administration (NASA)* June 2026 – August 2026

- **Supported the Artemis program** at Kennedy Space Center by developing **Moon Tycoon**, a **3D simulation of the lunar surface**. Engineers can explore and place robots anywhere on the moon with extremely realistic detail to test In-Situ Resource Utilization (ISRU) equipment such as **rovers, drills, vehicles, landers, and lunar modules**
- Developed custom driving controls and **cylinder wheel physics**, overcoming PhysX limitations for lunar terrain
- Generated lunar surface detail and used real-time data to calculate sun position and local lighting conditions
- Built system to select coordinates on a lunar globe and teleport there with **accuracy to the meter**
- Optimized software framerate (FPS) by 3x and reduced loading time from 20 seconds to 2 seconds

**President & Founder** | *Game Dev Club Club: a club for game dev clubs*

March 2026 – Present

- Founded a club for game dev clubs, hosting over **100 universities** and **1600+ members** from multiple countries
- Organized the **Club Club Jam Jam**, which broke the record as the **largest online student-run collegiate game jam** in the world. It was officially featured by Itch.io and **received 470 participants and over 150 entries**
- Jam was judged by **YouTubers BenBonk and Jerbob** (100k+ subscribers), and **ScoreSpace** (games publisher)

**Undergraduate Research Assistant** | *Diane Chi Fund, University of Pennsylvania*

May 2026 – September 2026

- Authored a novel **mesh-based simulation of natural plant growth and flower development**
- Simulated biological differential growth and cell differentiation to procedurally grow shoots, leaves, branches
- Simulated flow of auxin growth hormone in plant cells to influence growth rate, cell type, and leaf venation
- **Implemented the half-edge mesh data structure on the GPU**, complete with edge splitting, Delaunay edge relaxation, cloth-sim-like self-collision detection, turgor pressure, and Laplacian smoothing
- Created custom node-based graph editor to visually define plant genotypes, cell types, and growth parameters

**Co-President** | *UPenn Games Research and Development Environment (UPGRADE) Club*

May 2025 – August 2026

- Organized events, led meetings, produced and presented workshops, taught game dev to new members
- Led small groups to create **13 unique minigames** and directed club-wide projects to **publish 2 games on Steam**

## SIDE QUESTS / PROJECTS

**Space Minecraft: a spherical voxel engine** | C++, OpenGL, GLSL, Qt Creator

November 2025 – December 2025

- Minecraft... in space! Place blocks on Saturn's rings, circumnavigate Earth, mine into the Sun, and inhabit Mars
- Created spherization and distortion reduction methods to convert flat terrain into **voxelized spherical planets**
- Built **deferred rendering engine from scratch** and post-processing effects like screen-space ambient occlusion
- Built **physics engine from scratch**: gravity/orbits, kinematics, collision, and Outer Wilds-like velocity matching
- Created **physically based sunsets & volumetric atmospheres** using Rayleigh/Mie light scattering, Henyey-Greenstein phase functions, and ozone layer simulation
- Built **LOD system** that runs quadtree compression on 3D planet surfaces, achieving a **40x** performance increase

**A Bear Game** | *Unity, Blender, Photoshop, C#, Shader Graph, VFX Graph, DaVinci Resolve*      *December 2025 – Present*

- In-development **3D platformer where you play as a bear** who was fired from his job and must save the world
- **Complex state machine with buffered & directional jumps/air jumps**, dynamic ledge grabs, wall runs, rail slides, ground pounds, combat, **60+ unique movement animations**, and **custom physics rigid body collision**
- Developed many editor tools incl. a custom node-based dialogue editor and GPU-instanced grass painters
- Explore the open world: beaches, cities, forests; drive on roads, talk to people, solve puzzles, and buy donuts
- Produced devlog videos on Instagram that have **garnered over 25k views**

**Physically Based Fluid Simulation & Renderer** | *Unity, C#, HLSL*      *June 2025 – August 2025*

- Simulated Newtonian fluids using **Smoothed Particle Hydrodynamics** and the Navier-Stokes equations
- Optimized runtime with **3D spatial hashing** and bitonic merge sort to support **260,000+ particles at 200 FPS**
- Rendered with **real-time raymarching** and physically based light rays to simulate **refraction, reflection, Fresnel**

**Flocks of Fish** | *Unity, C#, Shader Graph, VFX Graph, HLSL*      *May 2025 – June 2025*

- Simulated the emergent behavior of flocking animals, like birds and schools of fish, using the Boids algorithm
- Implemented GPU acceleration to achieve stable performance of **20,000+ flocking fish at 200 FPS**
- Hop in a toon submarine and explore a deep ocean of flocking fish, sharks, whales, coral, and bubbles

**Game Jams** | *Unity, Unreal, Blender, C#, Blueprints, FL Studio, Aseprite, Figma*      *February 2021 – Present*

- Participated in **15+ game jams**, from puzzle platformers to a pen spinning game with Tony Hawk-like combos
- Placed **25th Overall / 3,506 entries** in Juniper Dev Jam and **10th Overall / 737 entries** in Brackeys Game Jam
- Became the first 2x consecutive winner of the University of Pennsylvania x Drexel (UPGRADE) Game Jam
- Honed **skills across game design**, incl. coding, 2D/3D art, animation, polish, level design, music, SFX, UI/UX
- Collaborated with many game developer friends; **shared many fun experiences and nights making games!**

SKILL TREE

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**Programming Languages:** C#, C++, C, Java, Python, HLSL, GLSL, JavaScript, TypeScript, HTML, CSS, SQL, OCaml

**Platforms/Tools:** Unity, Unreal, OpenGL, UE Blueprints, Shader Graph, Houdini, Blender, ZBrush, Maya, DaVinci Resolve, After Effects, Audacity, FL Studio, Aseprite, Figma, Photoshop, Paint.NET, React, Tailwind, Github, VSCode